Jiaxin (Jeremiah) Pan

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Research Interests

Physics undergraduate working on AI for Science, generative sampling, and interpretable crystal descriptors. Interested in reliable machine learning, language-model reasoning, and music-related AI; combines scientific computing with piano and choral training.

Education

2023–2027 **B.Sc. in Physics (with Honours) — First Class Honours expected**
The Chinese University of Hong Kong, Shenzhen
GPA: 3.61/4.0. Four-year degree with a required research thesis. Admitted to Economics in 2023; completed Year 1 foundation courses before transferring to Physics in 2024.

2026 (Spring) **Visiting Student**
University of Auckland · New Zealand
Completed coursework in Physics, Computer Science, and AI, including COMPSCI 361 and COMPSCI 220.

2025 (Summer) **Summer Visiting Student**
University of California, Berkeley Extension · USA
Completed coursework in physics and neuroscience.

Current Research

Ongoing **Data-Free Generative Sampling of the Two-Dimensional XY Model**
Undergraduate Thesis, CUHK-Shenzhen
Supervised by Prof. Kai Zhou. Provisional portfolio title.

- Contributed to adapting Stochastic Path Sampling (SPS) to periodic XY angles using energy and forward/backward path probabilities without equilibrium training data; evaluated paths with path-space independence Metropolis–Hastings (MH) correction.
- Completed reproducible Python/PyTorch training-budget experiments across three seeds and three generation-step settings on Slurm-managed GPUs; used Wolff sampling as an independent validation reference.
- Evaluated physical statistics, effective sample size per sampling second, and rejection streaks; found consistent early gains and seed-dependent later changes. Scope: 4×4, one temperature; no established thermodynamic critical behavior or advantage over Wolff.

Ongoing **Interpretable Crystal Representations for AI-Assisted Materials Design**
HSSRLM — AI for Science — Working paper
Hierarchical crystal descriptors with T5 (Text-to-Text Transfer Transformer) bidirectional text/descriptor infrastructure; broader data validation and model evaluation ongoing.

- Contributed to evidence-aware CIF (Crystallographic Information File)-to-descriptor preprocessing preserving provenance and known/inferred/missing information.
- Contributed to candidate validation, node/linker/topology connectivity checks, explicit failure reporting, and descriptor-level duplicate tracking.
- Contributed to limited structure-equivalence audits on three publicly sourced metal–organic frameworks, examining information lost by coarse descriptors.

Descriptors omit full atomistic structures; rule checks do not prove physical validity. Co-first author; manuscript in preparation for planned submission to the Annual Meeting of the Association for Computational Linguistics (ACL). Not yet submitted.

Additional Research Experience

2026 (Spring) **Remote Research Assistant**
INSAIT · Bulgaria / Remote
Under Dr. Jinjin Gu: technical analysis of large language model reasoning and post-training, including Surgical Post-Training (SPOT), knowledge retention, and regularization; prepared research syntheses and contributed to seminars.

2025 (Fall) **Guided Research on Van der Waals Integration and 2D Electronics**
CUHK-Shenzhen, Prof. Qi Qian's Lab
Hands-on fabrication and characterization of MoS_2 devices using dry transfer and van der Waals integration; studied metal–semiconductor contacts and polarity modulation. Conducted literature review and prepared an end-of-term report.

2025 (Spring) **Computational Physics Course Project**
CUHK-Shenzhen
Group study of the quantum harmonic oscillator using finite-difference discretization and matrix eigenvalue methods; compared numerical and analytical results, examined convergence, and synthesized the final report in LaTeX. Technical note on Zenodo (2026).

Manuscript and Conference

Submitted **Mathematics self confidence and STEM major choice in an undergraduate survey**
Reinvention: an International Journal of Undergraduate Research
Sole author; awaiting editorial assessment. Earlier version presented by invitation at the CUHK-Shenzhen Undergraduate Research Conference (URC2026), 31 January 2026. Earlier course paper (Zenodo DOI). Preprint posted on SSRN.

Teaching

Oct–Dec 2025 **Undergraduate Student Teaching Fellow (USTF)**
CUHK-Shenzhen School of Science and Engineering
Provided tutoring through the Student Help Room, a form of USTF service, for CSC1005 (Introduction to Computer Engineering), CSC1001 (Introduction to Computer Science), and PHY1001 (Mechanics), 23 October–12 December 2025.

Musical Training and Performance

Piano Passed the Chinese Musicians Association amateur piano Grade 8 examination.

Choral singing Tenor in CUHK-SZ Chorus (Fall 2026), Auckland University Student Choir (Spring 2026), and School of Music Choir, CUHK-SZ (choir class, Fall 2025). Performed in two public Auckland concerts and *Winter Carols* at CUHK-SZ. Performance record; concert recordings.

Honours and Certification

2023–2026 Full-Tuition Scholarship, CUHK-Shenzhen; awarded based on the 2023 Gaokao.

2024–2026 Dean's List: School of Management and Economics (AY2023–2024); School of Science and Engineering (AY2024–2025 and AY2025–2026). The 2026 certificate is pending issuance.

2024–2025 Shaw Overcoming Adversity Award, CUHK-Shenzhen Shaw College.

2025 HCIA-AI: Huawei Certified ICT Associate — Artificial Intelligence; valid through December 2028.

Skills and Languages

Computing Python, PyTorch, NumPy, SciPy, Matplotlib, Git, LaTeX; GPU experiment workflows through Slurm.

Languages English: TOEFL iBT 100 (October 2025); Mandarin: native. Choral pronunciation training in German, Cantonese, Latin, Maori, and Japanese.